ESE 3207 One-Shot Notes

Solar Photovoltaic Systems · KUET · past papers 2019, 2020, 2021, 2022, 2023 · answers from the teachers' material (Rahaman sir: Parts I-III; Muhit sir: lectures L1-L11) and the books they follow (Smets et al. *Solar Energy*, Messenger *Photovoltaic Systems Engineering*)

One-shot notes: ESE 3207 Solar Photovoltaic Systems

Constants

q 1.602e-19 C; h 6.626e-34; c 3e8; hc = 1240 eV·nm; kT/q 0.0258 V; ε_{Si} = 11.7 × 8.854e-14 F/cm; Si E_g 1.12 eV, n_i 1.5e10, N_C 3.22e19, N_V 1.83e19 cm^{-3}. STC 1000 W/m^2, AM1.5, 25 °C.

Doping and bands (A1)

P (V) donor → n, E_D below E_C, E_F up. B (III) acceptor → p, E_A above E_V, E_F down. Purpose: conductivity, majority carrier, junction field. $np = n_i^2$.

Fermi = 50% occupancy; DOS $g(E) \propto \sqrt{(E - E_C)}$; Poisson $d^2\psi/dx^2 = -\rho/\varepsilon$ (field from charge); continuity: change = flow + G − R; $L = \sqrt{(D\tau)}$.

Direct: same k, photon only, α high (GaAs, CdTe). Indirect: needs phonon (Si, Ge). Drift $J = qn\mu E$; diffusion $J = qD\, dn/dx$; $D/\mu = kT/q$.

Quantum: quantised energy ($E = h\nu$), wave-particle duality, uncertainty, Pauli/allowed states.

Recombination (A2)

Radiative (photon, direct gap); SRH (trap in gap, phonons, defects; dominant in Si); Auger (third carrier, heavy doping); surface (dangling bonds; cure SiN_x/Al_2O_3, BSF). G = R in equilibrium.

2021 1c: $\Phi_0 = P\lambda/hc$ = 2.52e21; ×0.62 = 1.56e21; αd = 333 so all absorbed; **J = 250 A/m^2** (book 248).

Junction (A3)

p-n: diffusion → ions → field → light pairs diffuse to edge → field separates → contacts collect. p-i-n: field across i layer, drift device (a-Si, perovskite). Schottky: $\varphi_m > \varphi_s$ (n) barrier, rectifying; ohmic: no barrier / heavy doping.

$V_{bi} = kT/q \ln(N_A N_D/n_i^2)$; $W = \sqrt{(2\varepsilon(V_{bi} - V)(1/N_A + 1/N_D)/q)}$.

2022 3d: **0.81 V, 0.325 µm, 5e4 V/cm**. 2021 3b: p = 1e16, n = 2.25e4; n = 1e18, p = 225. 2020 3b: L_n 25.7, L_p 22.4, W 0.11 µm, **$I_L = qAG(L_n + W + L_p)$ = 11.6 A**.

Parameters and circuit (A4)

I_{sc}, $V_{oc} = kT/q \ln(I_L/I_0 + 1)$, MPP, $FF = V_m I_m/V_{oc} I_{sc}$, $\eta = P_m/(GA)$, EQE/IQE.

Real: $I = I_L - I_0(e^{q(V+IR_s)/nkT} - 1) - (V + IR_s)/R_{sh}$; two-diode n = 1 and 2. R_s (bulk, emitter, fingers, contacts) tilts near V_{oc}; R_{sh} (edge leaks, defects) tilts near I_{sc}; both cut FF.

2020 1a: **FF 0.781, η 13.0%**. 2021 5d: **FF 0.724, η 17.8%** (P_{in} 280.6 W).

Losses and limits (A5)

Spectral: non-absorption (~19%), thermalisation (~33%). Optical: shading, reflection, parasitic, transmission. Electrical: recombination, R_s, R_{sh}; voltage factor, FF.

Rules: texture + ARC, fewer/finer fingers, back reflector; passivate, BSF, selective emitter; heavy doping under contacts. BSF: p/p^+ field repels electrons from rear, $+V_{oc}$, +red response.

Thermodynamic 86.8% (conc.), 68% (1 sun, ∞ junctions); Carnot 95%. SQ: one junction, radiative only → **33.7% at 1.34 eV**; Si practical 29.4%.

Si wafer and cell (A6)

Quartzite: > 98% SiO_2, lumpy, stable, cheap. $SiO_2 + 2C \rightarrow Si + 2CO$ (arc 1900 °C, MG 98%) → $Si + 3HCl \rightarrow HSiCl_3 + H_2$ → distil → Siemens 1100 °C → poly 9N → CZ/FZ ingot → square → wire saw → lap, etch, polish.

Cell: damage etch, texture (KOH), $POCl_3$ 850 °C, PSG off (HF), edge isolate, SiN_x PECVD, print Ag/Al, co-fire (Al-BSF), test.

Litho: clean, spin resist, soft bake, align + UV expose, develop, hard bake, etch, strip. TCO: ITO/FTO/AZO, E_g > 3 eV, > 80% T, conductive front electrode for thin films.

Thin film and advanced (A7, A8)

Criteria: E_g 1.1-1.7, high α, long L, dopable, stable, abundant, non-toxic, cheap. III-V: GaAs, InP, GaInP, AlGaAs; thin because direct gap (1-3 µm), epitaxial; 29%/47%.

CdTe: glass / TCO / n-CdS / p-CdTe (1.45 eV, $CdCl_2$) / p^+ back contact. a-Si:H: H passivates dangling bonds; p-i-n drift; Staebler-Wronski.

MJ: GaInP 1.86 / GaAs 1.42 / Ge 0.67, series + tunnel junctions; challenges: bandgap, current, lattice matching, tunnel junction, cost. Lattice 5.65 Å all three. Perovskite ABX_3 ($MAPbI_3$): tunable direct gap, solution processed, > 26%; n-i-p FTO/TiO_2/pvk/spiro/Au; unstable, Pb.

Gen I wafer Si; II thin film; III beyond SQ: MJ, up/down conversion, MEG, hot carrier, intermediate band, perovskite, DSSC, OPV.

Module and array (B1)

Cell 0.5 V; module = 36/60/72 in series (Wp at STC); array = strings in parallel. Series V adds; parallel I adds. $I_{sc} \propto G$ and area; $V_{oc} \propto \ln G$; V_{oc} −2.3 mV/°C; P −0.45%/°C. Face south, tilt ≈ latitude (23°).

2022 5c: 75 Wp → **68.25, 64.9, 61.5 W**. L3: 19 cells (0.765 V at 60 °C); 6 parallel. Cooling: air gap, PV/T, water, low-γ tech.

Shading (B2)

Series current = weakest cell; shaded cell reverse biased → hot spot. Bypass diode: parallel, reverse, per 18-24 cells. Blocking diode: series, stops night discharge and reverse string current. Mismatch: manufacturing/operating; current (series) vs voltage (parallel). Fix: bypass, binning, multi-MPPT, optimisers, micro-inverters, global MPPT.

Controller, MPPT, DC-DC (B3)

Functions: overcharge, deep discharge (LVD), reverse current, load control. Set points (12 V): VR 14.4, ARV 13.2 (VRH), LVD 11.5, LRV 12.6 (LVDH). Shunt: shorts array + blocking diode. Series: opens line.

PWM: bulk → pulses narrow as SOC rises → float. MPPT: DC-DC + algorithm, impedance matching R = V_m/I_m, +10-30%. P&O: step, keep if P up, else reverse (oscillates). IncCond: dI/dV = −I/V at MPP (stops, fast). Buck $V_o = DV_i$; boost $V_i/(1 - D)$.

SG 1.265 full; autonomy days; lm/W; THD. L5: 5 A, 20 A; Ah 83.3%, Wh 69%.

Inverters (B4)

Square (THD ~45%), modified sine (~20%), pure sine (< 5%, SPWM + LC). Standalone, grid-tied, bimodal. Cut THD: SPWM, filters, multilevel, SHE.

2022 6c: **264 W out, 36 W loss**. 2021 6c: 420 W → input 466.7 W, 38.9 A DC → **600 VA 12 V pure sine**. L9: 4000 W, 16.7 A.

Systems and net metering (B5)

Off-grid: PV → CC → battery → inverter → load. On-grid: PV → GTI → DB + net meter → grid. With storage: hybrid inverter, backup, self-consumption. Hybrid: PV + diesel/wind + ECC.

Grid conditions: V, f, phase sync, THD < 5%, DC < 0.5%, pf, anti-islanding 2 s, 5 min reconnect, lockable switch, meter.

Net metering: pay import − export. BD 2018: 3-phase consumer, ≤ 70% sanctioned load, own premises, no dues, approved inverter + bi-directional meter, agreement.

Pump direct: cheap, no battery, day only. With battery: anytime, costly, losses.

SHS sizing (B6)

E → PV = E/(η·PSH) → Ah = E·days/(V·DOD) → CC = 1.25I → inverter = 1.25P → switch 1.25I → A = 2ρLI/ΔV.

2023/2019: E **3280 Wh**, load 1060 W; 4373/5 = 875 → **900 Wp**; **1822 Ah** (10 × 200); CC **100 A**; inverter **1500 VA**; switch **125 A**; cables 25, 50, 16 mm^2 (lengths assumed).

2022 6d: E **3180**; **700 Wp**; **1060 Ah**; CC 80 A; inverter 800 VA-1 kVA.

Economics and ecology (B7)

PBT = money back (5-10 yr); EPBT = energy back (1-3 yr). Energy yield ratio 10-15. LCOE = $\Sigma(I + M + F)/(1 + r)^t \div \Sigma E/(1 + r)^t$; grid parity, socket parity.

Costs: capital, O&M, replacement, fuel, finance, salvage. x = 1.03/1.10; P_a 11.50, P_{a1} 10.76.

2020 6c: PV **$6,726** (3000 + 300 + 900 + 648 + 466 + 336 + 1076); generator **$25,727** (fuel $1.25/gal assumed).

CO_2: PV 20-50 g/kWh vs coal ~900; from Si purification. Recycle frame, glass, Si, Ag, Cu; dumping leaches Pb, Cd.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
a-Si:H	hydrogenated amorphous silicon
Ah	ampere-hour
ALCC	annualised life-cycle cost
AM	air mass
ARC	anti-reflection coating
ARV	array reconnect voltage
AZO	aluminium-doped zinc oxide
BOS	balance of system
BSF	back surface field
CB	conduction band
CIGS	copper indium gallium diselenide
CZ	Czochralski
CZTS	copper zinc tin sulfide
DOD	depth of discharge
EPBT	energy payback time
EQE	external quantum efficiency
ETL	electron transport layer
EVA	ethylene vinyl acetate (encapsulant)
FA	formamidinium
FF	fill factor
FTO	fluorine-doped tin oxide
FZ	float zone
HIT	heterojunction with intrinsic thin layer
HTL	hole transport layer
HVD	high-voltage disconnect
IBC	interdigitated back contact
IEEE	Institute of Electrical and Electronics Engineers
IncCond	incremental conductance
IQE	internal quantum efficiency
ITO	indium tin oxide
LCA	life-cycle assessment
LCC	life-cycle cost
LCOE	levelized cost of electricity
LRV	load reconnect voltage
LVD	low-voltage disconnect
LVDH	low-voltage disconnect hysteresis

MA	methylammonium
MBE	molecular beam epitaxy
MEG	multiple exciton generation
MG-Si	metallurgical-grade silicon
MOCVD	metal-organic chemical vapour deposition
MPP	maximum power point
MPPT	maximum power point tracking (tracker)
nc-Si:H	hydrogenated nanocrystalline silicon
NOCT	nominal operating cell temperature
P&O	perturb and observe
PBT	payback time
PECVD	plasma-enhanced chemical vapour deposition
PERC	passivated emitter and rear cell
PERL	passivated emitter, rear locally diffused
PID	potential-induced degradation
PSG	phosphosilicate glass
PSH	peak sun hours
PV	photovoltaic
PV/T	photovoltaic-thermal (hybrid collector)
PWM	pulse width modulation
SHS	solar home system
SOC	state of charge
SPV	solar photovoltaic
SPWM	sinusoidal pulse width modulation
SQ	Shockley-Queisser
SREDA	Sustainable and Renewable Energy Development Authority (Bangladesh)
SRH	Shockley-Read-Hall (trap-assisted recombination)
STC	standard test conditions (1000 W/m^2, AM1.5, 25 °C)
TCO	transparent conducting oxide
TCS	trichlorosilane ($HSiCl_3$)
THD	total harmonic distortion
TOPCon	tunnel oxide passivated contact
VA	volt-ampere
VB	valence band
VFD	variable frequency drive
VR	voltage regulation (set point)

VRH	voltage regulation hysteresis
Wp	watt-peak (power at STC)